

Anteater Poker
Version 1.0

EECS 22L: Software Engineering Project

University of California, Irvine

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T3

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Table of Contents

- Glossary
- 1. Online Poker
 - 1.1 User Scenario
 - 1.2 Goals
 - 1.3 Features
- 2. Installation
 - 2.1 System Requirements
 - 2.2 Unpacking & Configuration
 - 2.3 Uninstalling
- 3. Program Functions & Features
 - 3.1 Client & Server Communication
 - 3.2 Dealer Choice & Card Distribution
 - 3.3 Poker Game Integration
- 4. Copyright / Error Messages / Index

Glossary

Program Technical Terms

- **Client** - Program that sends requests to the server. Includes an interactive graphical user interface (GUI) and an automated Poker Bot.
- **Server** - Program that processes the clients' requests, maintains player records, cards, points, and game status.
- **Graphical User Interface (GUI)** - Constructed using GTK 3.0 libraries with hierarchical widgets and an event-based main control loop.
- **Socket** - Endpoint for networked inter-process communication using TCP/IP protocols.
- **Event-Based Main Control Loop** - An architectural pattern where the program waits for and dispatches user events (i.e. button clicks) rather than following a static procedural path.
- **GTK 3.0** - The specific cross-platform widget toolkit used to create the program's GUI.
- **Hierarchical Widgets** - The structural organization of GUI elements where parent containers (like windows or boxes) hold child elements (like buttons or labels).
- **TCP/IP (Transmission Control / Internet Protocol)** - The suite of communication protocols used to connect hosts on the Internet, facilitating reliable data transfer between the client and server.

Administrative / Build Terms

- **Gtar (GNU Tar)** - The tool used to create or extract "tar-balls", which are compressed archive files containing source code.
- **Make / Makefile** - A build automation tool that automatically builds executable programs and libraries from source code by reading instructions in a "Makefile".
- **Source Archive** - A compressed (e.g., .tar.gz) containing the raw code, documentation, and assets required to compile the software.
- **Unit Test** - A software testing method by which individual modules of source code are tested to determine whether they are fit for use before system integration.

Glossary ... continued

Game-Related Terms

- **Ante** - A small mandatory bet contributed by all players before a hand is dealt to create an initial pot.
- **Anteater Card** - A custom face card added to the deck in addition to the standard Jack, Queen, and King.
- **Bet** - The act of placing a wager into the pot, which subsequent players must at least match to remain in the hand.
- **Call** - To match the current highest bet or raise placed by another player.
- **Check** - To decline to bet when it is the player's turn and no prior bet has been made in the current round.
- **Community Cards** - Cards dealt face-up on the table that are shared by all players to form their best hand.
- **Fold** - To discard a hand and forfeit interest in the current pot instead of matching a bet.
- **Hand** - The best combination of cards a player can form using their private cards and the community cards.
- **Pot** - The total amount of points or bets accumulated during a single game, awarded to the winner(s).
- **Raise** - To increase the size of the existing bet in the current round, forcing other players to match the new amount or fold.
- **Wild Card (Joker)** - A special card that can be used to represent any other card to improve a player's hand ranking.

1. Online Poker

1.1 Usage Scenario

Users can play a fun Poker game with friends remotely. Players are able to meet with their friends online using a video or voice conferencing tool (i.e. Discord, Zoom, or etc.) and play cards using the Anteater Poker application on their computer.



[Figure 1: Conceptualized Poker Game GUI]

1.2 Goals

To provide a robust networked card game using a client-server architecture and an event-driven graphical user interface.

1.3 Features

- Players will be able to enter their name and game configuration choices, which includes their player seat and password.
- Players can view their own cards, the table state, other players' statuses, and the current points.
- Interactive actions, including check, raise, call, and fold, alongside placing bets.
- Integration of a Poker Bot that can act as a computer player to fill available seats and take actions automatically.
- Custom Anteater rule extensions, such as the inclusion of an "Anteater" face card alongside Jacks, Queens, and Kings, or specialized actions cards for exchanging cards and gaining points.

2. Installation

2.1 System Requirements

This application is specifically designed for the Linux operating system. You will require a standard C/C++ build environment (GCC), Make, and the GTK 3.0 libraries for GUI rendering. Network compatibility for TCP/IP communication is required.

2.2 Unpacking & Configuration

To unpack the software from the source archive and compile the binaries, execute the following commands from top to bottom:

- Extract the source archive: `gtar xvzf Poker_V1.0.tar.gz`
- Navigate to the directory: `cd Poker_V1.0`
- Compile all components: `make all`
- (Optional) Run tests: `make test`

2.3 Uninstalling

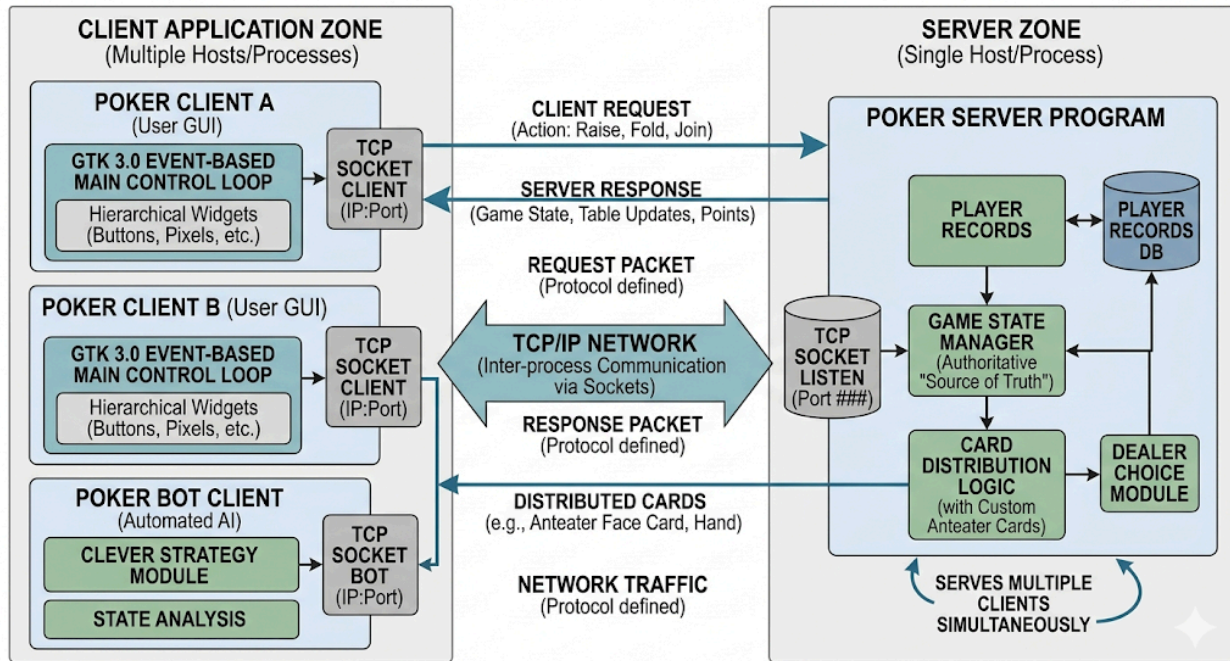
To remove the compiled binaries and clean the directory structure, execute the following command:

- `make clean`

3. Program Functions & Features

3.1 Client & Server Communication

The system uses a networked inter-process communication model. The Poker client sends requests to the Poker server with TCP/IP sockets. The server processes these requests and serves multiple connected client applications simultaneously, providing both data and game flow control.



[Figure 2: Client-Server Communications Diagram]

3.2 Dealer Choice & Card Distribution

The system uses a networked inter-process communication model. The Poker client sends requests to the Poker server with TCP/IP sockets. The server processes these requests and serves multiple connected client applications simultaneously, providing both data and game flow control.

- **Client Application Zone:** Includes multiple hosts/processes such as Poker Client A/B (User GUI) and the Poker Bot Client (Automated AI).
- **Server Zone:** A single host/process that runs the Poker Server Program, functioning as the authoritative "Source of Truth".
- **Network Traffic:** Defined protocols facilitate the exchange of Request Packets (e.g., Raise, Fold, Join) and Response Packets (e.g., Game State, Table Updates, Points).

3.3 Poker Game Integration

Players input their requested actions (e.g., check, raise, call, fold) through the event-based GUI. The server updates the overall game status and points. If humans are unavailable, the Poker Bot, functioning as a non-interactive client program with a clever strategy, will automatically analyze the game state and place bets to ensure continuous gameplay.

4. Copyright / Error Messages / Index

Copyright

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Error Messages:

- **“Connection Refused”** - The Poker server is not running or the IP address is incorrect.
- **“GTK-WARNING: cannot open display”** - Ensure your X11 forwarding is enabled if running remotely.
- **“Segmentation Fault”** - Usually indicates a memory leak in the C code or a socket buffer overflow.

Index:

A	Section	Page
Ante (mandatory bet)	Glossary	4
Anteater Card (Custom Face Card)	Glossary/3.2	4,7
Anteater Poker (App overview)	1.0	5
B		
Bet	Glossary	4
Build Env (GCC/Make)	2.1	6
C		
Call	Glossary/1.3	4,5
Card Distribution	3.2	7
Check	Glossary/1.3	3
Client - Server Architecture	3.1	7
Community Cards	Glossary	4

Configuration	1.3/2.2	5,6
Connection Refused	4	9
Copyright	4	9

D

Dealer Choice Mod	3.2	7
Discord/Zoom	1.1	5

E

Error	4	9
Event-Based Main	Glossary/1.3	3,8
Extraction	2.2	6

F

Fold	Glossary/1.3	7
Features	1.3	5

G

Game State	3.1	7
GTK	Glossary/2.1	3,6
GUI	Glossary/1.3	3,5
Goals	1.2	5
Glossary	Glossary	3-4

H

Hand	Glossary	4
Hierarchical Widgets	Glossary	3

I

Installation	2.0	6
---------------------	------------	----------

J

Joker	Glossary/3.2	4,7
--------------	---------------------	------------

L

Linux	2.1	6
--------------	------------	----------

M

Make/Makefile	Glossary/2.2	3,6
----------------------	---------------------	------------

Make all	2.2	6
-----------------	------------	----------

Make clean	2.3	6
-------------------	------------	----------

Make test	2.2	6
------------------	------------	----------

N

Network Reqs	2.1	6
---------------------	------------	----------

O

Online Poker	1.0	5
---------------------	------------	----------

P

Password	1.3	5
-----------------	------------	----------

Player Records	3.1	7
-----------------------	------------	----------

Player seat	1.3	5
--------------------	------------	----------

Poker bot	Glossary/1.3/3.3	3,5,8
------------------	-------------------------	--------------

Pot	Glossary	4
------------	-----------------	----------

Program features	3.0	7
-------------------------	------------	----------

R

Raise	Glossary/1.3	4,5
--------------	---------------------	------------

Remote Play	1.1	5
--------------------	------------	----------

S

Segmentation	4	9
---------------------	----------	----------

Server	Glossary/3.1	3,7
---------------	---------------------	------------

Socket	Glossary/3.1	3/7
---------------	---------------------	------------

Source Archive	Glossary/2.2	3,6
-----------------------	---------------------	------------

System Requirements	2.1	6
----------------------------	------------	----------

T

Table State	1.3	5
--------------------	------------	----------

TCP	Glossary/3.1	3,7
------------	---------------------	------------

Team members	Cover	1
---------------------	--------------	----------

U

Unit test	Glossary/2.2	3,6
------------------	---------------------	------------

Uninstalling	2.3	6
---------------------	------------	----------

Unpacking	2.2	6
------------------	------------	----------

V

Version 1.0	Cover	1
--------------------	--------------	----------

X

X11 Forwarding	4	9
-----------------------	----------	----------